

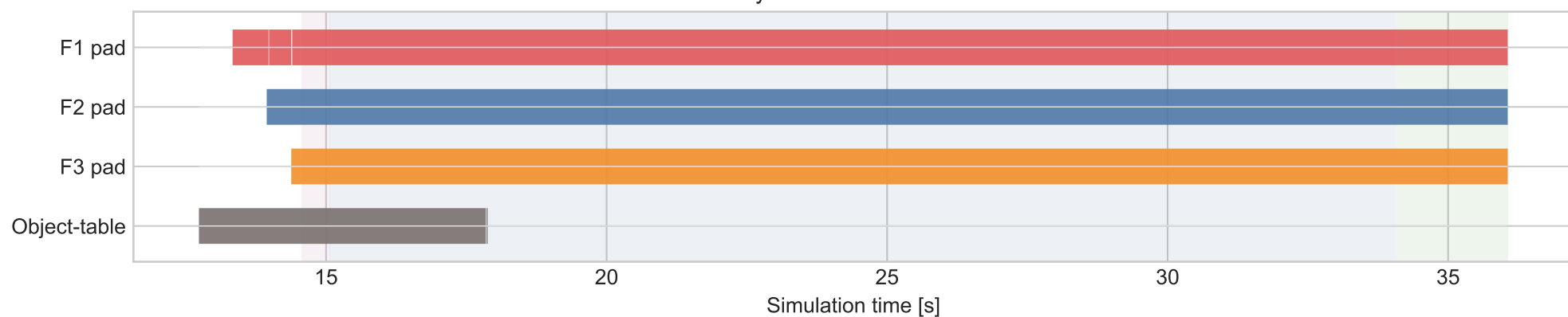
Frozen-scene nominal grasp: measured motion and native contact evidence



GPU-native per-step equivalent normal force (unfiltered)



Physical contact records



lift=50.889 mm, hold=2.0 s, slip=0.709 mm — deterministic simulation only